Higher National Diploma in Information Technology

First Year, First Semester Examination - 2023

HNDIT 1022-Web Design

Q1.

1. Define the term **Internet Protocol (IP)** . [02 Marks]
2. List two main purposes of using the **internet** in daily life . [04 Marks]
3. Explain the concept of **DNS** and its role in web browsing . [06 Marks]
4. Fill in the blanks with suitable words from the list given below . [08 Marks]

   (TCP/IP, Domain Name, HTTP, FTP, URL, Bandwidth, HTTPS, Router, ISP, Cloud Storage)

   a. _____ is the secure version of HTTP used for encrypted data transfer .

   b. _____ is a hardware device that directs data across networks .

   c. A unique name that identifies a website on the Internet is called a _____ .

   d. _____ is a protocol used for the reliable transmission of data packets across a network .

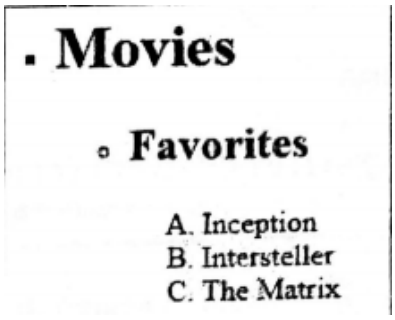
[Total 20 Marks]

Q2.

1. What does **HTML** stand for? [02 Marks]
2. Name four main structural elements in an HTML document . [04 Marks]
3. Write the HTML code to create the following list of favorite movies . [06 Marks]

   Hint: You may find the following information helpful for your answer:

   - Movies - H1
   - Favorites - H2
   - Inception, Interstellar, The Matrix - default Size



4. Fill in the blanks in the following spaces by applying your HTML knowledge . [08 Marks]

   a. The tag for adding a hyperlink is <_____>.

   b. An image can be embedded using the <_____> tag.

   c. The _____ attribute is used to specify the URL for an <a> tag .

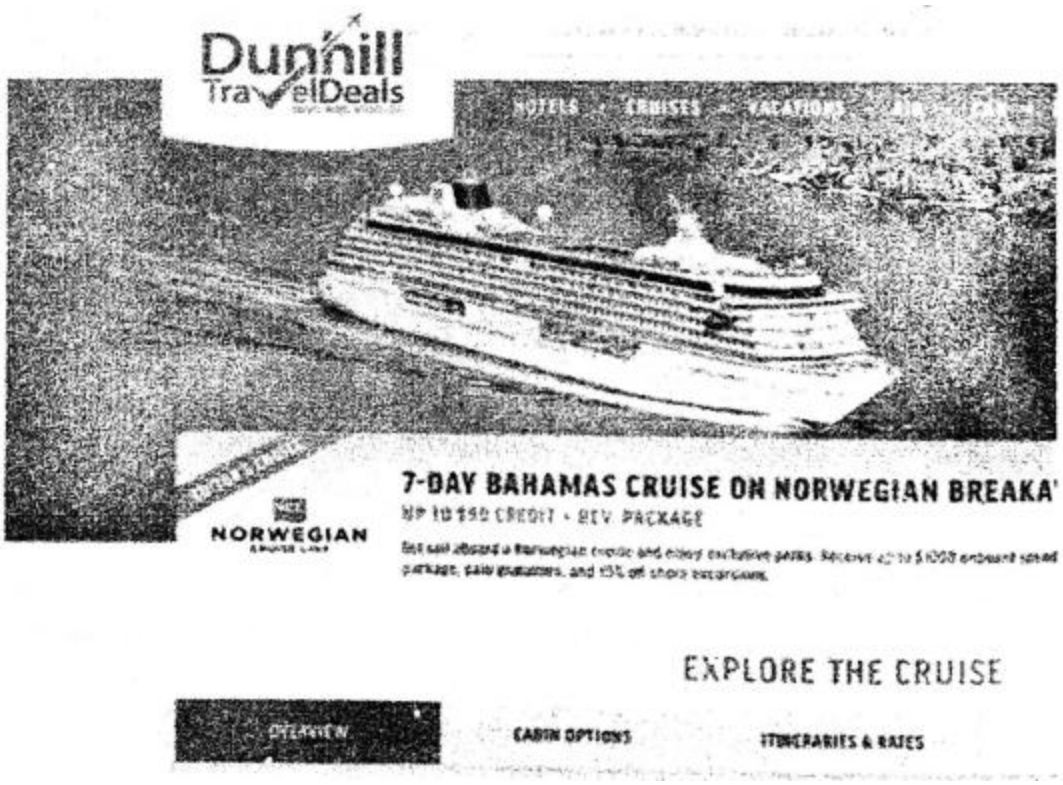
   d. HTML documents typically start with <!DOCTYPE _____>.

[Total 20 Marks]

Q3.

1. Define a **wireframe** in web design . [02 Marks]
2. Why are wireframes essential in the design process? [04 Marks]
3. List three common elements included in a wireframe . [06 Marks]
4. Sketch the **wireframe design** for the given web page . [08 Marks]

[Total 20 Marks]



Q4.

1. What is **CSS** used for in web development? [02 Marks]
2. List and describe two benefits of using **External CSS** in web design . [04 Marks]
3. Write **CSS code** to set all paragraph text to **14-point Arial font** . [06 Marks]
4. Fill in the blanks in the CSS properties and values . [08 Marks]

   a. To specify font style as italic, the _____ property is used .

   b. _____ is used to set the background image for an element .

   c. To align text to the center, the _____ property is used .

   d. The property to specify border color is _____ .

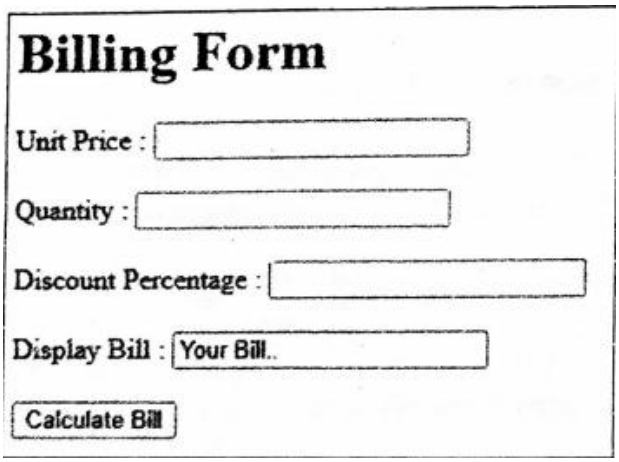
   [Total 20 Marks]

Q5.

1. List two common uses of **JavaScript** in web development . [02 Marks]
2. Name two rules for defining variables in JavaScript . [04 Marks]
3. What is the **output** of the following JavaScript code snippet? [06 Marks]

```
<script type="text/javascript">// Define the course variable

var course = 'HNDA';

document.write("Entering switch block<br />");

switch (course) {

   case 'HNDIT':document.write("Higher National Diploma in Information Technology<br
/>");

      break;

   case 'HNDA':document.write("Higher National Diploma in Accountancy<br />");

   case 'HNDTHM':document.write("Higher National Diploma in Tourism and Hospitality
Management <br />");

   case 'HNDM':document.write("Higher National Diploma in Management<br />");

      break;

   case 'HNDE': document.write("Higher National Diploma in English<br />");

      break;

   default: document.write("Unknown course<br />");

}

document.write("Exiting switch block");

</script>
```

4. a. Write the required **HTML code** to display the following form . [04 Marks]



b. Write a JavaScript function to calculate the bill and display it in the appropriate text box . [04 Marks]

Hint: When you click the Calculate Bill button, the bill will be displayed in the text box.

Formula: Bill = Unit Price * Quantity - (Unit Price * Quantity * Discount Percentage * 1/100)

[Total 20 Marks]

Q6.

1. Define **web hosting** . [02 Marks]
2. List two reasons why web hosting service providers offer **uptime guarantees** . [04 Marks]
3. What are the three types of web hosting plans? Briefly describe each . [06 Marks]
4. Fill in the blanks related to web hosting features: [08 Marks]

    a. _____ is a control panel for managing hosting services .

    b. _____ provides security for the website .

    c. _____ is the amount of data transferred between a website and its users .

    d. _____ ensures that a website's data is saved regularly to prevent data loss .

    [Total 20 Marks]